Refinements to the Kuznetsov Formula

for Laplace–Beltrami Operators

Merrick S. Chang

October 1, 2026

Outline

1. Setup and Statement of Results

2. Proof of Theorem A: The Codimension-One Case

3. Proof of Theorem B: Measure-Zero Looping Directions

Setup and Statement of Results

Spectral Asymptotics: Weyl's Law

- (M, g) compact Riemannian, $\dim M = n$. Spectrum of $-\Delta_g$:

$$0 = \lambda_0^2 \leq \lambda_1^2 \leq \cdots$$

- Eigenvalue counting function:

$$N(\lambda) = \#\{j : \lambda_j \leq \lambda\} = (2\pi)^{-n}\omega_n \operatorname{Vol}_g(M)\, \lambda^n + O(\lambda^{n-1})$$

  where ω_n is the volume of the unit ball in $\mathbb{R}^n$.

- **Duistermaat–Guillemin [3] / Ivrii [6]:** remainder improves to $o(\lambda^{n-1})$ *iff* periodic geodesics form a measure-zero subset of T^*M.

- $H \subseteq M$ closed embedded submanifold, $\dim H = d$.
- Period integral $\int_H e_j \, dV_H$ measures concentration of e_j along H.
- Roots in analytic number theory: Kuznetsov [7], Bruggeman [2] — Fourier coefficients of automorphic forms vs. Kloosterman sums.

Zelditch's [12] one-term Kuznetsov formula

$$N_H(\lambda) := \sum_{\lambda_j \leq \lambda} \left| \int_H e_j \, dV_H \right|^2 = C_{H,M} \, \lambda^{n-d} + O(\lambda^{n-d-1})$$

So what if we put the two together?

Wyman–Xi: A Two-Term Formula

Let $\dot{N}^*H = N^*H \setminus 0$, Φ_t the geodesic flow on $T^*M \setminus 0$.

$$\mathcal{T}_0 = \{t \in \mathbb{R} : \Phi_t(x,\xi) \in \dot{N}^*H \text{ for some } (x,\xi) \in \dot{N}^*H\} \cup \{0\}$$

$$\Sigma_t = \{(x,\xi) \in \dot{N}^*H : \Phi_t(x,\xi) \in \dot{N}^*H,\ d\Phi_t \text{ an isomorphism}\}$$

Proposition (Wyman–Xi [11])

If $S\Sigma_t$ has measure zero in SN^*H for every $t \in \mathcal{T}_0$, and $\mathcal{T}_0$ is countable, then

$$N_H(\lambda) = C_{H,M}\,\lambda^{n-d} + o(\lambda^{n-d-1}) + C.$$

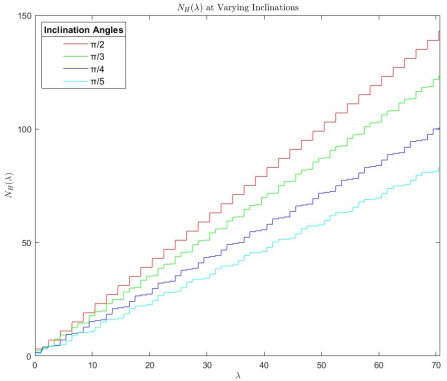
Can We Improve Wyman-Xi?

The previous formula does not perfectly parallel the Duistermaat–Guillemin law at two key points

1. **The residual constant** C which is not required in the original law
2. **Countability hypothesis** which is not an exact analogue for Duistermaat–Guillemin

Numerical Evidence for $C = 0$

- Wyman–Xi tested $M = T^2$, $H = S^1$.
- We tested $M = S^2$, $H =$ longitudinal circles of varying inclination.
- In every case, the linear fit to $N_H(\lambda)$ passes through the origin.



Suggestive, but not a proof — this motivates Theorem B.

Definition of Set of Looping Directions

Define the set of *looping directions*

$$\left\{(x,\xi) \in S\dot{N}^*H \;\middle|\; \Phi_t(x,\xi) \in S\dot{N}^*H \text{ for some } t > 0\right\}.$$

This Talk: Two Refinements

Theorem A (codimension one)

If $\dim H = \dim M - 1$ and $N_H(\lambda) = C_{H,M}\lambda + C + o(1)$, then in fact

$$N_H(\lambda) = C_{H,M}\lambda + o(1).$$

The stray constant is always 0.

Theorem B (measure-zero looping directions)

If the set of looping directions in SN^*H has measure zero, then

$$N_H(\lambda) = C_{H,M}\lambda^{n-d} + o(\lambda^{n-d-1}).$$

No countability assumption on $\mathcal{T}_0$ *is needed- a more direct fit.*

Proof of Theorem A: The Codimension-One Case

Strategy

- **Goal:** assuming $N_H(\lambda) = C_{H,M}\lambda + C + o(1)$, show $C = 0$.

Strategy

- **Goal:** assuming $N_H(\lambda) = C_{H,M}\lambda + C + o(1)$, show $C = 0$.
- **Method:** Heat kernel test

Strategy

- **Goal:** assuming $N_H(\lambda) = C_{H,M}\lambda + C + o(1)$, show $C = 0$.
- **Method:** Heat kernel test
- The constant term C in N_H corresponds to a specific power of t on one side

Step 1: From N_H to a Heat Trace

Integrating by parts against $e^{-t\lambda^2}$ and inserting the assumed asymptotic,

$$\int_{-\infty}^{\infty} e^{-t\lambda^2}\, dN_H(\lambda) = \int_0^{\infty} 2t\, e^{-t\lambda^2} \left(C_{H,M}\lambda^2 + C\lambda + r(\lambda)\lambda\right) d\lambda, \quad r(\lambda) = o(1).$$

Evaluating the first two terms explicitly:

$$\int_0^{\infty} 2t\, e^{-t\lambda^2} (C_{H,M}\lambda^2 + C\lambda)\, d\lambda = \frac{C_{H,M}\sqrt{\pi}}{2\sqrt{t}} + C.$$

Step 2: The Heat Trace, Computed Directly

Expanding N_H in eigenfunctions and summing,

$$\int_{-\infty}^{\infty} e^{-t\lambda^2}\, dN_H(\lambda) = \sum_{\lambda_j \le \lambda} e^{-t\lambda_j^2} \int_H \int_H e_j(x)\overline{e_j(y)}\, dV_H(x)\, dV_H(y)$$
$$= \int_H \int_H e^{t\Delta_g}(x,y)\, dV_H(x)\, dV_H(y).$$

Reformulated goal

Show that the double integral of the heat kernel over $H \times H$ has expansion

$$C_{H,M}\, t^{-1/2} + \big(\text{terms vanishing as } t \to 0\big),$$

i.e. **no constant (t^0) term.** Comparing with Step 1 forces $C = 0$.

The Heat Kernel, Locally

- Short-time asymptotics [5]: away from the cut locus,

$$p_M(t,x,y) \;\sim\; \left(\frac{1}{2\pi t}\right)^{n/2} e^{-\operatorname{dist}(x,y)^2/2t} \sum_{m\geq 0} H_m(x,y)\, t^m,$$

  with $H_0 > 0$, $H_0(x,x) = 1$.

- On a *compact* manifold

$$p_M(t,x,y) \leq \frac{C}{t^{(2n-1)/2}}\, e^{-\operatorname{dist}(x,y)^2/2t}$$

  globally, including the cutlocus.

Local Coordinates Adapted to H

- Geodesic normal coordinates + implicit function theorem: near $x \in H$, write

$$h = (x_1, \ldots, x_d, \varphi(x)),$$

with $\varphi(0) = 0$, $\nabla\varphi(0) = 0$.

- A direct computation with the metric $g = I + Y$, $Y = O(\|x\|^2)$, shows

$$dV_H = \det((dh)^T g(dh))dy' = \sqrt{1 + |f(y')|^2}\, dy',$$

for some f vanishing to order ≥ 2 at 0.

The Key Local Estimate

Lemma

On a sufficiently small U,

$$X = \frac{1}{(2\pi t)^{n/2}} \int_U e^{\frac{-|y'|^2 - |f(y')|^2}{2t}} \sum_{m \geq 0} H_m(0, y)\, t^m \sqrt{1 + |f(y')|^2}\, dy'$$
$$= C_U\, t^{-1/2} + o(1).$$

This is exactly what we need

The expansion has **no t^0 term**: the leading behavior is a pure half-integer power of t. Thus $C = 0$.

Proof Sketch of the Lemma

It is easy to check that

$$\frac{1}{(2\pi t)^{n/2}} \int_{\mathbb{R}^n} e^{\frac{-|y'|^2}{2t}}\, dy' = Ct^{-1/2} + o(1)$$

so break the calculation into a series of estimates to prove that

$$\frac{1}{(2\pi t)^{n/2}} \int_{\mathbb{R}^n} e^{\frac{-|y'|^2}{2t}}\, dy' - X = C\, t^{-1/2} + o(1).$$

Proof Sketch of the Lemma (Cont.)

A chain of estimates isolates the Gaussian leading term:

1. **Gaussian concentration:** outside $|y'| \leq 1$, all quantities are exponentially small in $1/t$.
2. **Curvature correction:** $\sqrt{1+|f(y')|^2} - 1 \leq C_0|y'|^2$, and after polar coordinates + the substitution $r = \sqrt{2tu}$, this contributes a vanishing integral $\int_0^{1/\sqrt{2t}} Cu^{(n+1)/2}e^{-u}\,du \to 0$.
3. **Warping correction:** $1 - e^{-|f(y')|^2/2t} \leq \min\{1, C|y'|^4/2t\}$; thus for $|y'| \lesssim t^{1/4}$, $e^{-|y'|^2/2t} - e^{\frac{-|y'|^2-|f(y')|^2}{2t}}$ is negligible.
4. **Higher heat coefficients:** each H_m, $m \geq 1$, contributes an extra power of t, hence vanishes relative to $t^{-1/2}$.

All four contributions vanish as $t \to 0$, leaving only $C_U t^{-1/2}$.

Proof of Theorem B: Measure-Zero Looping Directions

Strategy: A Tauberian Reduction

- Wyman–Xi already control an *averaged* version of N'_H (smoothed by a Schwartz function γ).
- We want the sharp asymptotic $N_H(\lambda) = C_{H,M}\lambda^{n-d} + o(\lambda^{n-d-1})$.
- We will borrow arguments from (Wyman 2017 [10]) which in turn borrow from the arguments of Ivrii 1980 ([3]).

Levitan's Tauberian Theorem

Proposition (Levitin, [9])

Let N be monotone, $N(\lambda) = 0$ for $\lambda \leq 0$, $N(\lambda) = o(\lambda^p)$. Let γ be even, Schwartz-class, positive, with $\hat{\gamma}(0) = 1$ and $\operatorname{supp} \hat{\gamma}$ compact. If

$$(N' * \gamma)(\lambda) = O(\lambda^\nu) \quad \text{and} \quad (N' * \rho)(\lambda) = o(\lambda^\nu)$$

for every ρ with $\hat{\rho} \in C_c^\infty(\mathbb{R})$, $\operatorname{supp} \hat{\rho} \subset (0, \infty)$, then

$$N(\lambda) = (N * \gamma)(\lambda) + o(\lambda^\nu).$$

Hypotheses of the Tauberian Theorem

Wyman–Xi supply the *first* hypothesis:

$$(N' * \gamma)(\lambda) = (2\pi)^{-n+d}\ \mathrm{Vol}(H)\ \mathrm{Vol}(S^{n-d-1})\,\lambda^{n-d-1} + O(\lambda^{n-d-3})$$

and thus

$$N_H(\lambda) = C_{H,M}\,\lambda^{n-d} + o(\lambda^{n-d-1}).$$

Our job: the *second* hypothesis, for every ρ supported away from 0.

Reduction to a Lemma

Lemma

For ρ with $\hat{\rho}$ compactly supported in $(0, \infty)$,

$$\sum_{\lambda_j \leq \lambda} \rho(\lambda - \lambda_j) \left| \int_H e_j \, dV_H \right|^2 \leq c \, \lambda^{n-d-1} + C \, \lambda^{n-d-2},$$

where c can be taken **arbitrarily small**.

This Lemma, together with Theorem A (to dispose of the constant when $n - d = 1$), gives Theorem B in full generality — with *no* countability assumption on $\mathcal{T}_0$.

Three Microlocal Tools (Wyman 2017)

1. A pseudodifferential partition of unity (localizing near H): operators $b, B \in \Psi^0_{\mathrm{cl}}(M)$ with

$$b(x,D) + B(x,D) = I, \qquad c = \int_{\mathbb{R}^d} \int_{S^{n-d-1}} |b(x', \omega)|^2 \, d\omega \, dx' < \epsilon,$$

and $\operatorname{ess\,supp} B$ avoids the "bad" set $\mathcal{L}_H([\delta, T])$ of directions that loop back to N^*H within time $[\delta, T]$.

2. Propagation of singularities: if $\mathrm{WF}(u) \cap \mathrm{WF}(v) = \emptyset$, then $t \mapsto \int_M e^{it\sqrt{-\Delta_g}} u \, \bar{v} \, dx$ is *smooth* in t.

3. A symbol estimate: Let $b(x,\xi)$ be a smooth function for $\xi \neq 0$, homogeneous of degree 0 in ξ. Define $b(x,D) \in \Psi^0_{\mathrm{cl}}(M)$ by

$$b(x,D)f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i\langle x-y,\xi\rangle} b(x,\xi)\, f(y)\, dy\, d\xi$$

for x, y, ξ expressed locally. Then

$$\sum_{\lambda_j \in [\lambda,\lambda+1]} \left| \int_H b\, e_j\, dV_H \right|^2 \leq C \left(\int_{\mathbb{R}^d} \int_{S^{n-d-1}} |b(x',\omega)|^2 h(x')^2\, d\omega\, dx' \right) \lambda^{n-d-1} + C_b\, \lambda^{n-d-2},$$

where C is independent of b and λ, and C_b is independent of λ but depends on b.

Proof of the Lemma

Denote the operator

$$X_\lambda(x,y) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{\rho}(t)\, e^{-it\lambda}\, e^{it\sqrt{-\Delta_g}}(x,y)\, dt.$$

Proof of the Lemma (Cont.)

Split the operator X_λ using the partition $b + B = I$ from the previous slide:

$$X_\lambda = \underbrace{BX_\lambda B^* + BX_\lambda b^* + bX_\lambda B^*}_{\text{negligible: } O(\lambda^{-N}) \text{ for all } N} + \underbrace{bX_\lambda b^*}_{\text{controlled by the symbol estimate}}$$

- Any term touched by B involves directions that provably do *not* loop back to N^*H within $[\delta, T]$ — Tool 2 makes these terms smooth in t, hence negligible to all orders.
- Only the fully-b term survives: it is supported on the (small, ϵ-mass) set of looping directions, and Tool 3 bounds it directly.

Proof of the Lemma (Cont.)

- If $(x,\xi) \in \mathrm{WF}(B^* V_H) \subset \operatorname{ess\,supp} B \cap N^* H$, then by construction

$$\Phi_t(x,\xi) \notin SN^*H \qquad \text{for all } \delta \le t \le T.$$

- By propagation of singularities,

$$\mathrm{WF}\left(e^{it\sqrt{-\Delta_g}} B^* V_H\right) = \Phi_t \ \mathrm{WF}(B^* V_H),$$

  which is therefore disjoint from $\mathrm{WF}(B^* V_H)$ itself on $[\delta, T]$.
- Hence $t \mapsto \int_M e^{it\sqrt{-\Delta_g}} (B^* V_H)\, \overline{B^* V_H}\, dx$ is smooth on $[\delta, T]$, and since $\operatorname{supp} \hat{\rho} \subset [\delta, T]$, integration by parts (in t) gains an arbitrary power of λ^{-1}.

The same argument (self-adjointness of X_λ) handles the two cross terms.

Proof of the Lemma (Cont.)

Only $bX_\lambda b^* = \sum_j \rho(\lambda - \lambda_j) \left| \int_H b\, e_j \, dV_H \right|^2$ remains.

- $X_\lambda(z)$ decays rapidly: $|X_\lambda(z)| \le C_N (1 + |z|)^{-N}$.
- Dominate by a dyadic step function $\sum_k a_k \mathbf{1}_{[k,k+1]}$, $a_k = \sup_{[k,k+1]} |\rho|$, and apply the symbol estimate on each shell:

$$\begin{aligned} \sum_j \rho(\lambda - \lambda_j) \left| \int_H b\, e_j \, dV_H \right|^2 &\le \sum_k a_k \Big(\epsilon \, (|\lambda + k| + 1)^{n-d-1} + C_b (|\lambda + k| + 1)^{n-d-2} \Big) \\ &\le \epsilon C \, \lambda^{n-d-1} + C_b \, \lambda^{n-d-2}. \end{aligned}$$

Conclusion of Theorem B

- $\epsilon > 0$ was arbitrary — the symbol mass of b near the looping set shrinks to 0 *precisely because that set has measure zero*.
- This proves the Lemma, hence the second Tauberian hypothesis, for every admissible ρ.

Putting it together

Levitan's theorem + Wyman–Xi's averaged estimate + the Lemma $\implies$

$$N_H(\lambda) = C_{H,M}\lambda^{n-d} + o(\lambda^{n-d-1}),$$

under the sole hypothesis that the looping directions have measure zero — and, via Theorem A, with **no** constant term when $n - d = 1$.

Acknowledgments

AI Acknowledgment

Claude was used to help draft parts of the introduction, clean up writing, prepare the slides, and check MATLAB code. All substantive mathematical results are the author's own work.

Thanks

To my advisor, Emmett Wyman, for his insights and for helping spot errors in the proof.

References i

[1] Vojislav G. Avakumović. **"Über die Eigenfunktionen auf geschlossenen Riemannschen Mannigfaltigkeiten".** In: *Mathematische Zeitschrift* 65 (1956), pp. 327–344.

[2] R. W. Bruggeman. **"Fourier coefficients of cusp forms".** In: *Inventiones Mathematicae* 45.1 (1978), pp. 1–18.

[3] J.J. Duistermaat and V.W. Guillemin. **"The Spectrum of Positive Elliptic Operators and Periodic Bicharacteristics.".** In: *Inventiones mathematicae* 29.1 (1975), pp. 39–80. URL: http://eudml.org/doc/142329.

[4] Lars Hörmander. **"The spectral function of an elliptic operator".** In: *Acta Mathematica* 121 (1968), pp. 193–218.

References ii

[5] Elton P Hsu. **Stochastic analysis on manifolds.** 38. American Mathematical Soc., 2002.

[6] Viktor Yakovlevich Ivrii. **"Second term of the spectral asymptotic expansion of the Laplace–Beltrami operator on manifolds with boundary".** In: *Functional Analysis and Its Applications* 14.2 (1980). Translated from *Funktsional'nyi Analiz i ego Prilozheniya* **14**:2 (1980), 25–34, pp. 98–106. DOI: 10.1007/BF01086550.

[7] N. V. Kuznetsov. **"Petersson's conjecture for cusp forms of weight zero and Linnik's conjecture. Sums of Kloosterman sums".** In: *Matematicheskii Sbornik (N.S.)* 111(153).3 (1980). English transl. in Math. USSR-Sb. **39** (1981), 299–342, pp. 334–383, 479.

[8] Boris M. Levitan. **"On the asymptotic behavior of the spectral function of a self-adjoint differential equation of the second order".** In: *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya* 16 (1952), pp. 325–352.

[9] Yu. Safarov and D. Vassiliev. **The asymptotic distribution of eigenvalues of partial differential operators.** en. Vol. 155. Translations of Mathematical Monographs. American Mathematical Society, Providence, RI, 1997. ISBN: 978-0-8218-4577-6. DOI: 10.1090/mmono/155.

[10] Emmett Wyman. **"Looping Directions and Integrals of Eigenfunctions over Submanifolds".** In: *The Journal of Geometric Analysis* 29 (June 2017). DOI: 10.1007/s12220-018-0039-x.

[11] Emmett Wyman and Yakun Xi. **"A Two Term Kuznecov Sum Formula".** In: *Communications in Mathematical Physics* 401.2 (May 2023), pp. 1127–1162. ISSN: 1432-0916. DOI: 10.1007/s00220-023-04667-z. URL: http://dx.doi.org/10.1007/s00220-023-04667-z.

[12] Steven Zelditch. **"Kuznecov Sum Formulae and Szegö Limit Formulae on Manifolds".** In: *Communications in Partial Differential Equations* 17.1-2 (1992), pp. 221–260. DOI: 10.1080/03605309208820840. eprint: https://doi.org/10.1080/03605309208820840. URL: https://doi.org/10.1080/03605309208820840.

Thank you.

Questions?